

Experience

Raytheon Technologies (RTX)

June 2023 - Present

Software Engineer II

Reusable Ground-Systems Visualization Framework

- Develop and maintain a reusable Angular 20 visualization framework used by multiple RTX programs to build browser-based satellite data displays.
- Reduced Largest Contentful Paint (LCP) by 55% in a production application built with the framework by introducing lazy loading for components previously loaded at startup.
- Led the framework's migration from Angular 18 to Angular 20, including adoption of zoneless change detection in its reference application and automated test environment.
- Design reusable visualization components and framework APIs that allow program teams to create specialized data displays without duplicating common infrastructure.

James Webb Space Telescope

- Modernized legacy operational displays originally developed for desktop systems in the 2000s as browser-based Angular applications used by JWST operators.
- Built end-to-end data-viewing features spanning the Angular frontend and Java-based Windows services, contributing backend changes to expose additional data and interfaces required by the web application.
- Proactively reworked communication between the application and its Web Workers, eliminating race conditions that caused displays to remain stuck loading or receive data inconsistently.
- Refactored a high-volume plotting component to reduce unnecessary chart updates and improve responsiveness while processing large bursts of JWST data.

Curve-10

August 2022 - May 2023

Software Engineer - Senior Capstone Project

- Led the frontend modernization of a legacy AngularJS application, rebuilding its primary interfaces in React.
- Integrated the React frontend with Node.js APIs and a MongoDB data layer while coordinating development with company engineers and a student team.

Raytheon Technologies (RTX)

May 2022 - August 2022

Software Engineering Intern

- Developed unit tests for an Angular/TypeScript application.
- Implemented Java services and PostgreSQL queries to supply satellite data to a Cesium-based frontend viewer.

Skills

Languages: TypeScript, JavaScript, Java, Python, SQL, C++, C#

Frontend: Angular, React, HTML, SCSS, RxJS

Backend & Data: Node.js, PostgreSQL, MongoDB, REST APIs

Testing & DevOps: Docker, GitLab CI/CD, Git, Jest

Education

University of Colorado Boulder, Bachelor of Science in Computer Science

August 2019 - May 2023